

CE1101



Geosphere



Biosphere



Hydrosphere



Atmosphere

Origin of Universe



https://www.youtube.com/watch?v=HdPzOWILrbE&ab_channel=NationalGeographic

Solar System



https://www.youtube.com/watch?v=libKVRa01L8&ab_channel=NationalGeographic

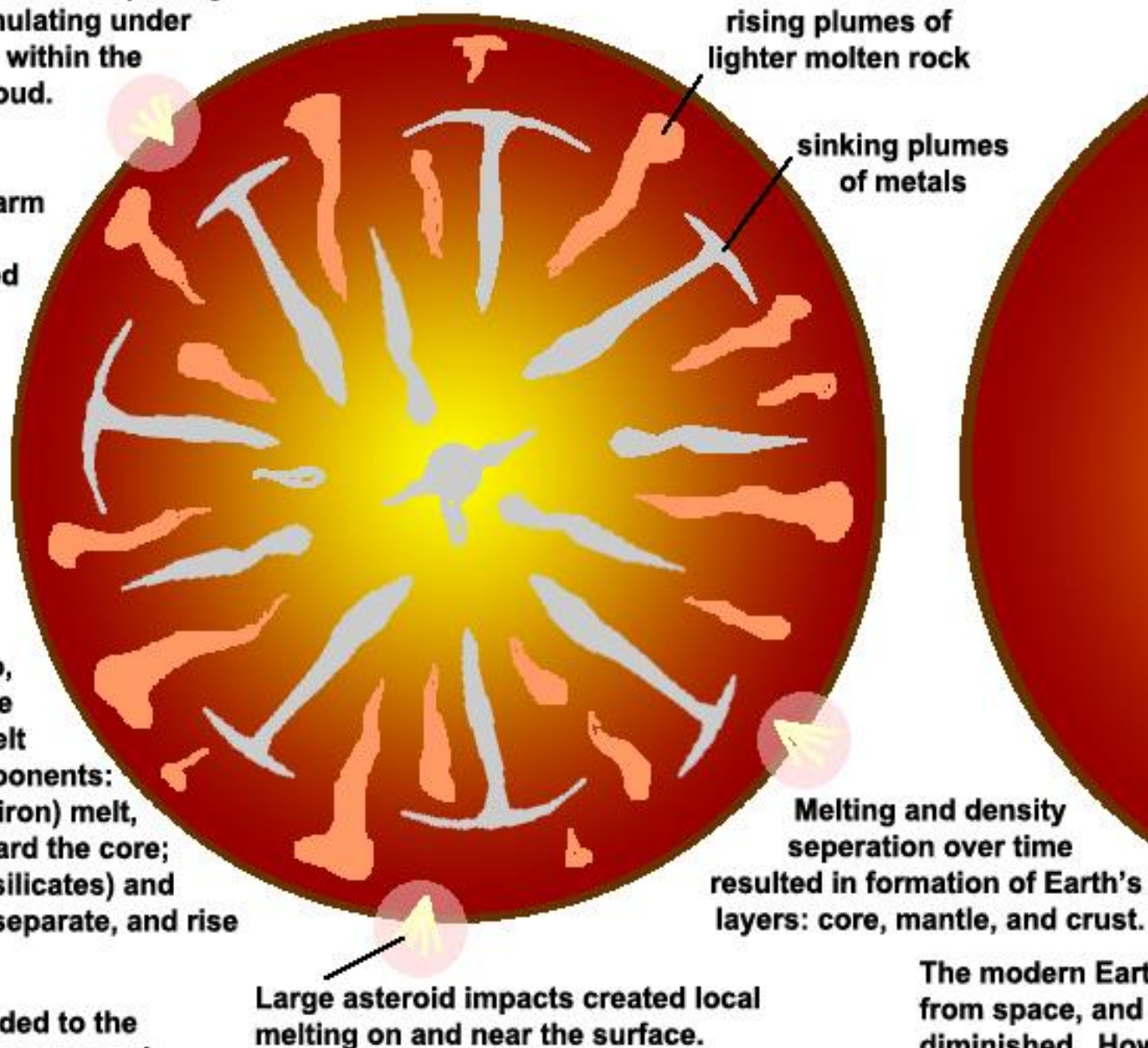
Early Earth formed from matter (mostly dust and gases) accumulating under gravitational attraction within the Sun's stellar nebula cloud.

As Earth grew in size, it gradually began to warm up. Sources of heat included heat generated by increasing compaction, radioactive decay of elements (mostly uranium, thorium, & potassium), and by heat generated by incoming meteor bombardment.

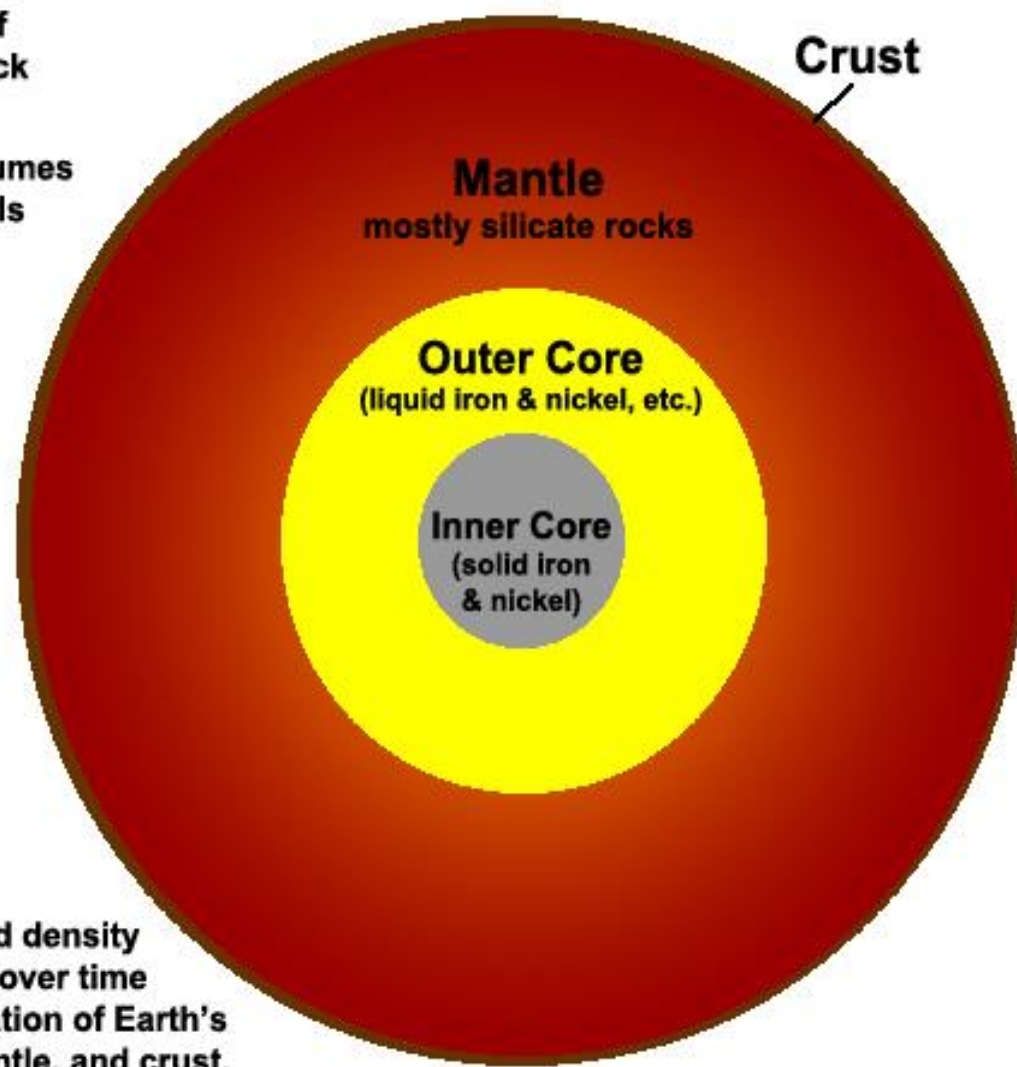
As early Earth heats up, temperatures rise to the point that rocks can melt and separate into components: denser metals (mostly iron) melt, separate, and sink toward the core; light minerals (mostly silicates) and dissolved gases melt, separate, and rise toward the surface.

Volcanic degassing added to the formation of the early oceans and atmosphere.

Early Earth



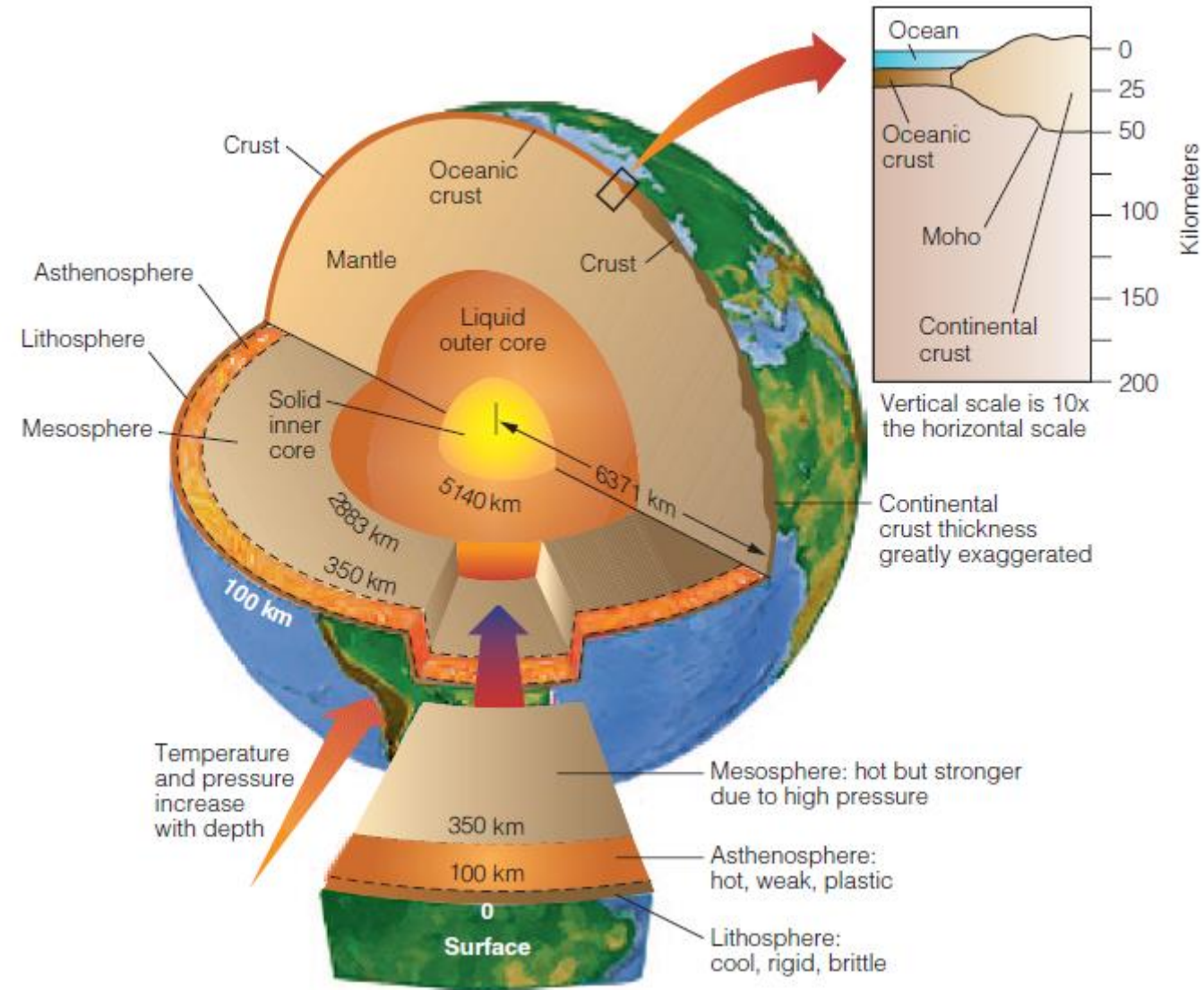
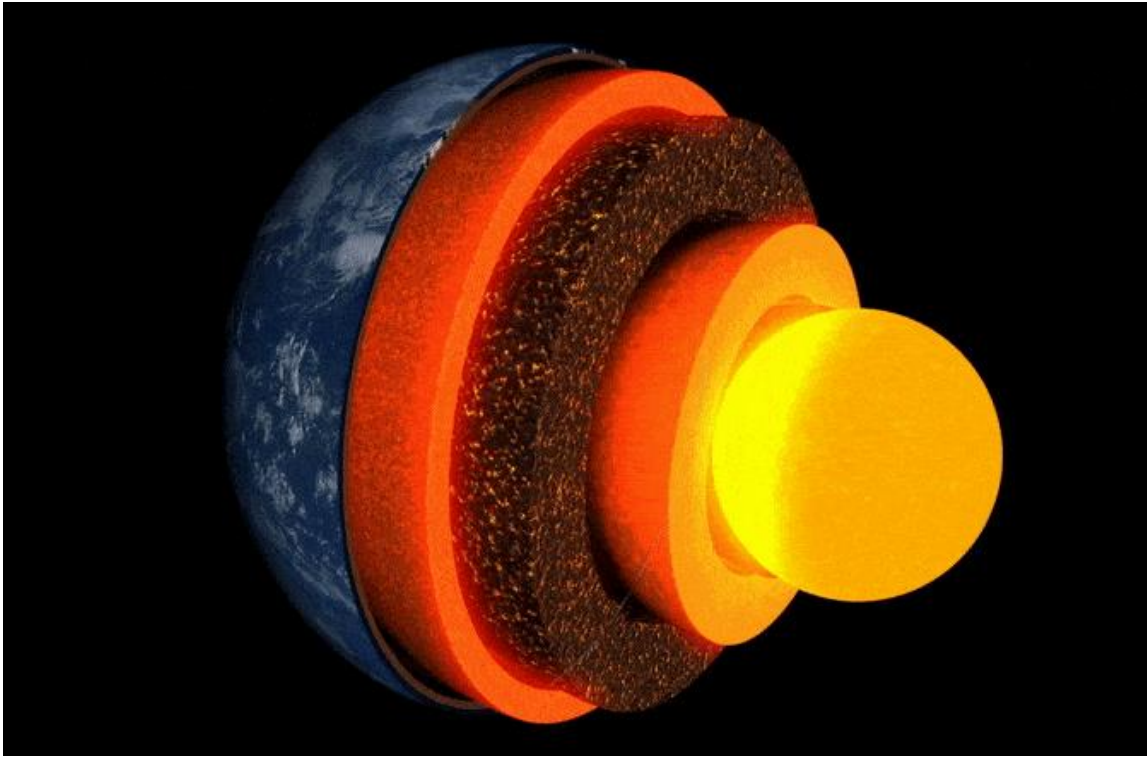
Modern Earth



Melting and density separation over time resulted in formation of Earth's layers: core, mantle, and crust.

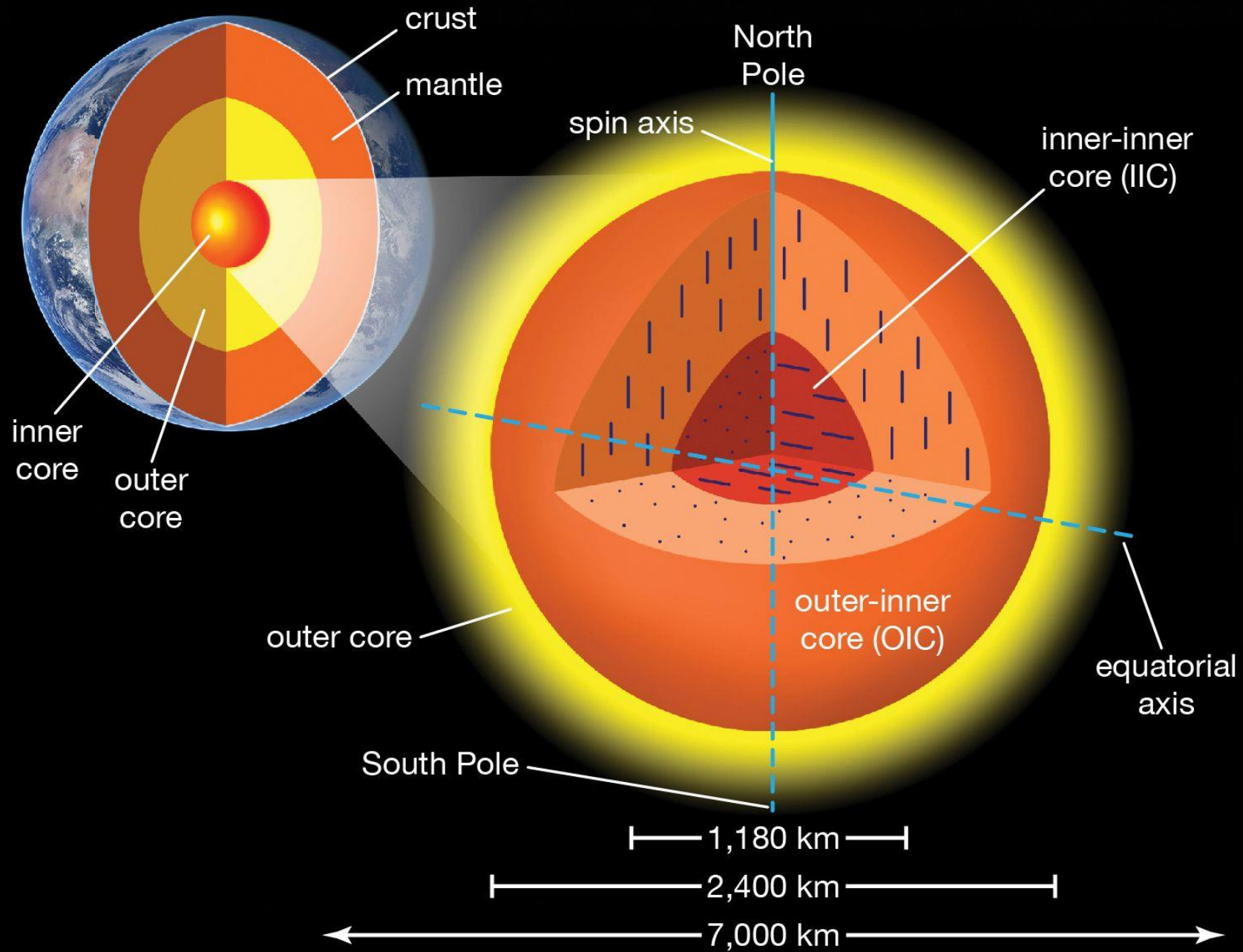
The modern Earth is no longer accumulating significant material from space, and the amount of heat from radioactive decay has diminished. However, the heat left over from the early period is still generating the heat that drives plate tectonics, volcanism, and the release of geothermal energy today.

Earth's interior



The Earth consist of several parts: a **core** (inner and outer), a **mantle**, and a **crust**

Earth's inner cores



Earth's interior - Crust

- The crust is the outermost solid shell of a rocky planet or moon, which is chemically distinct from the underlying mantle.
- The crust is mostly composed of relatively **low-density silicates minerals rich in aluminum**.
- The crust is a comparatively thin outer skin that ranges from about 2 miles (3 km) thick at the oceanic ridges to 40 miles (70 km) under some mountain belts.
- Gravity measurements show that the crust is separated into thin oceanic (3.0 gm/cc) and thicker continental crust (2.7 gm/cc).
- The crust is **more rigid** than the underlying mantle, it is brittle, and hosts earthquakes.



Earth's interior - Mantle

- The mantle is an inner layer of a terrestrial planet or other rocky body large enough to have differentiated in composition by density.
- On Earth, the mantle is a **highly viscous layer** between the crust and the outer core.
- Composed of higher-density **silicates rich in iron and magnesium** that extends to a depth of about 1800 miles (2900 km).
- Large portions can flow slowly and are near melting.



Earth's interior - Core

- The **core** is the innermost part of the earth is **believed** to be a **magnetic iron-nickel rich sphere** that consists of a 758 mile (1220 km) thick solid and a very dense **inner core** that is overlain by 1400 miles (2250 km) of dense molten material in the **outer core**.
- The **outer core is liquid**, and **heat convection** here creates currents in the liquid metal that generate **Earth's magnetic field**.

Other rocky planets and moons also have cores, mantles, and crusts, hydrosphere, and atmosphere. So far, only Earth is known to have a biosphere.

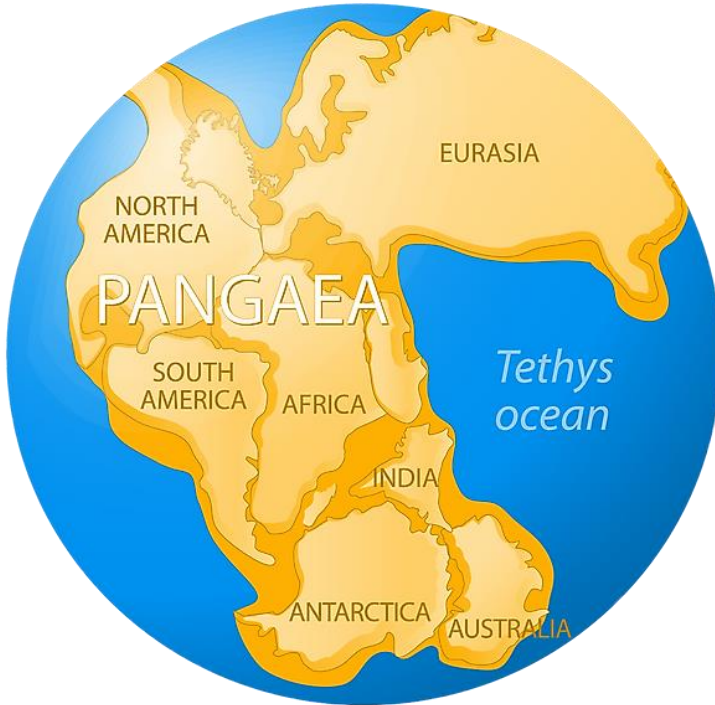


Tectonic cycle

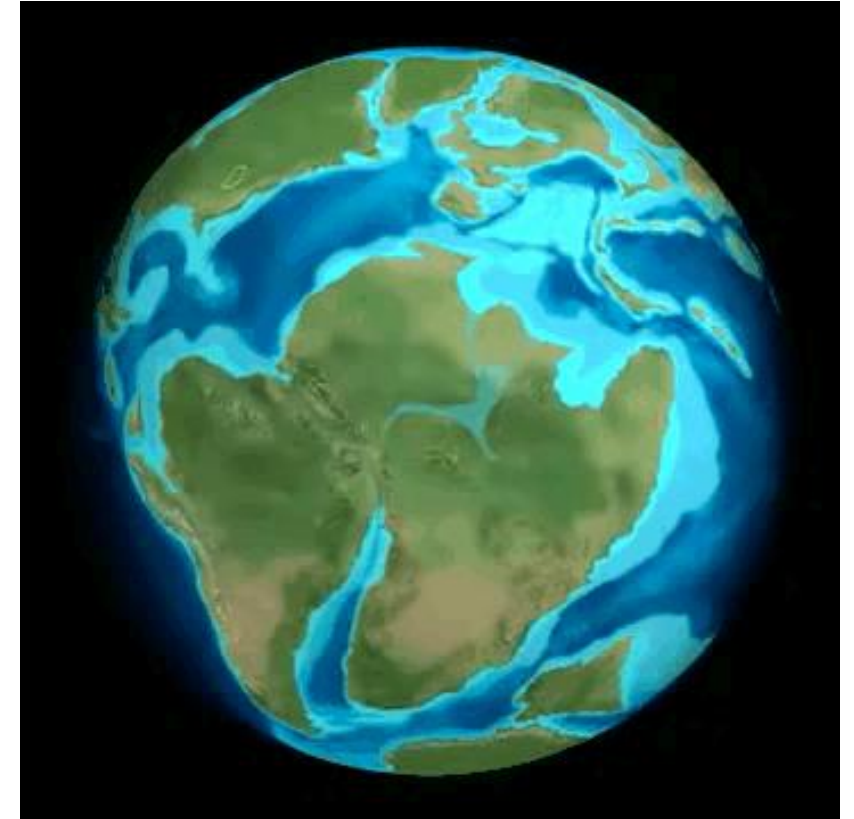
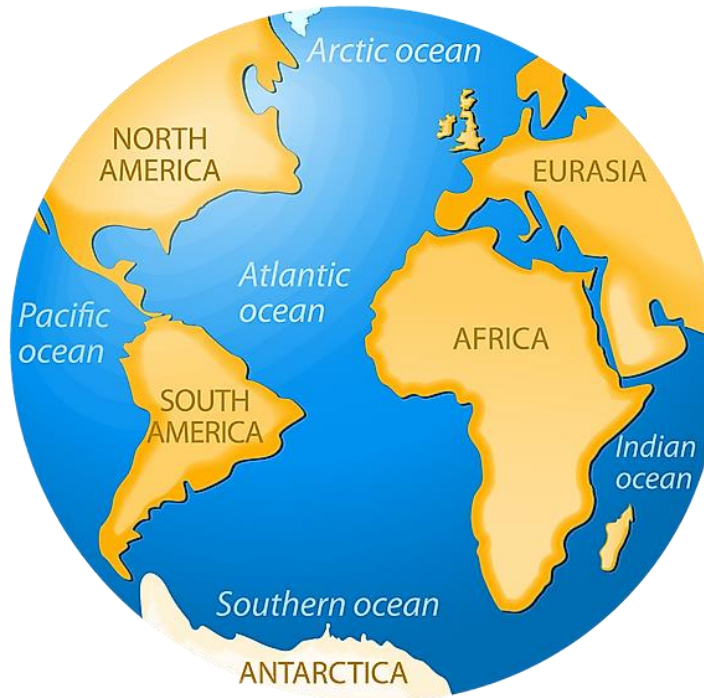


Continental drift theory

BEFORE



AFTER



Proposed by Alfred Wegener

- Single supercontinent landmass he named Pangea (meaning “all lands”).
- Northern part of Pangea is commonly called **Laurasia** and the southern part **Gondwana**.



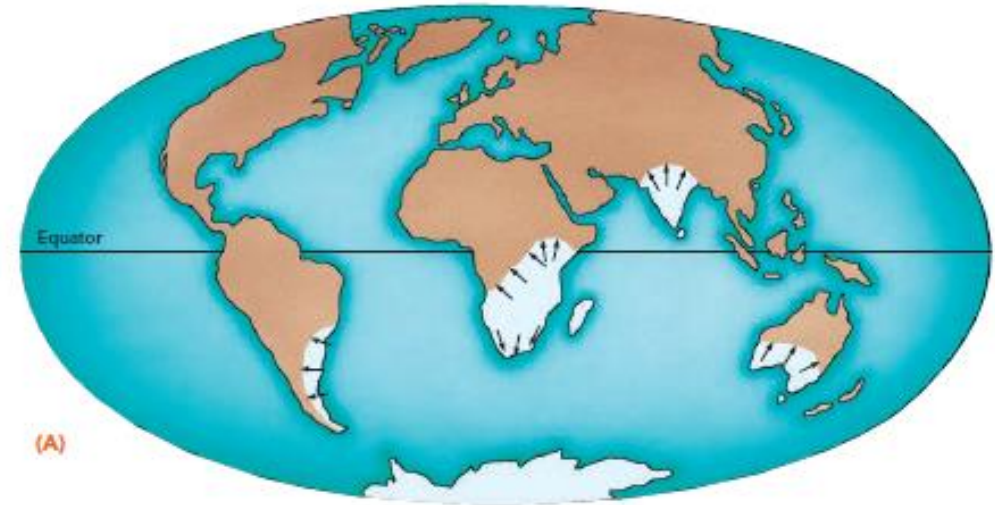
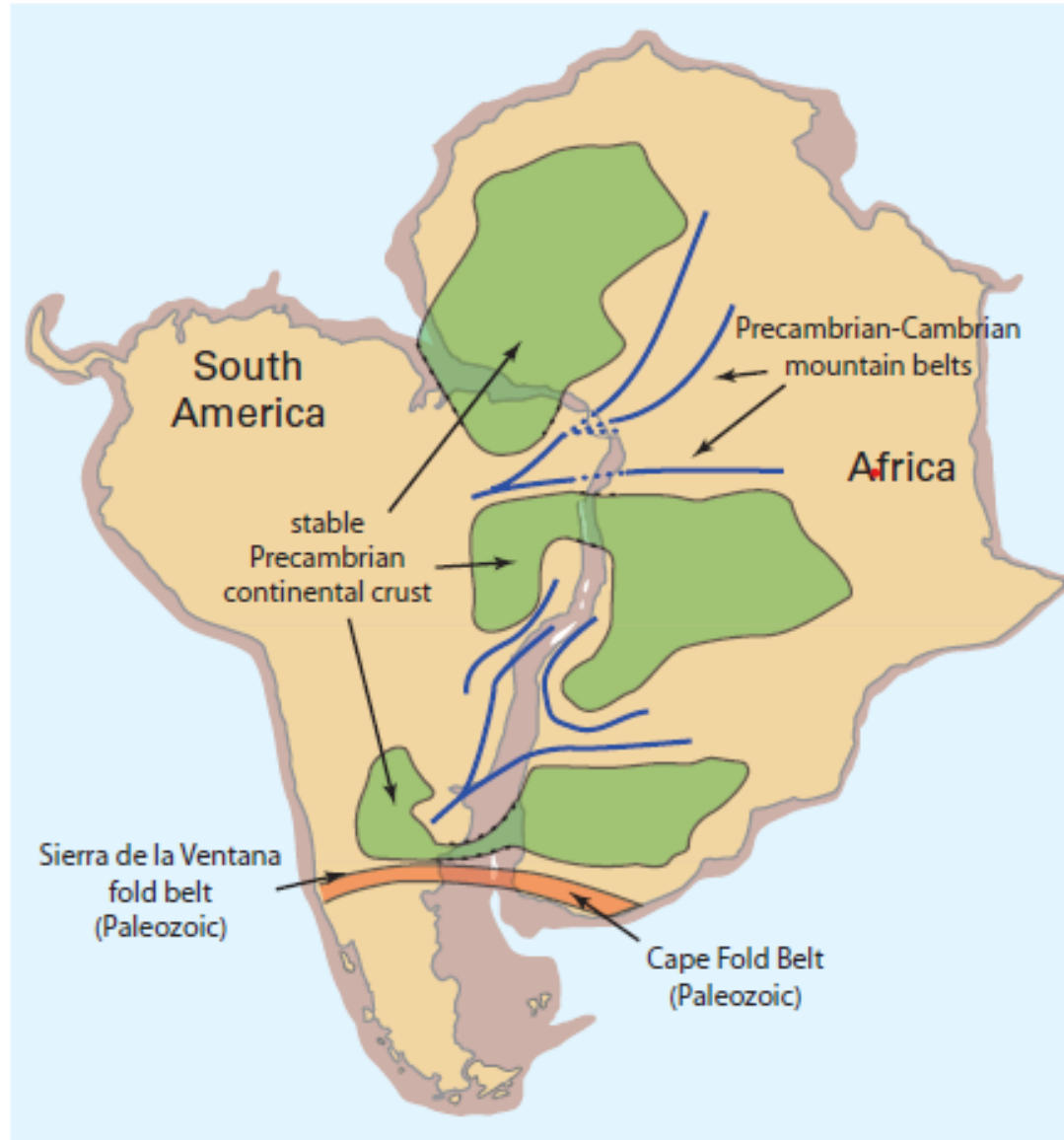
Continental drift theory

Wegener supported his theory with evidence such as the matching

- Coastlines of continents,
- Similar Plants, fossils and rock formations found on different continents.
- The presence of glacial deposits in regions that are now far from the poles.



Continental drift theory – Matching features

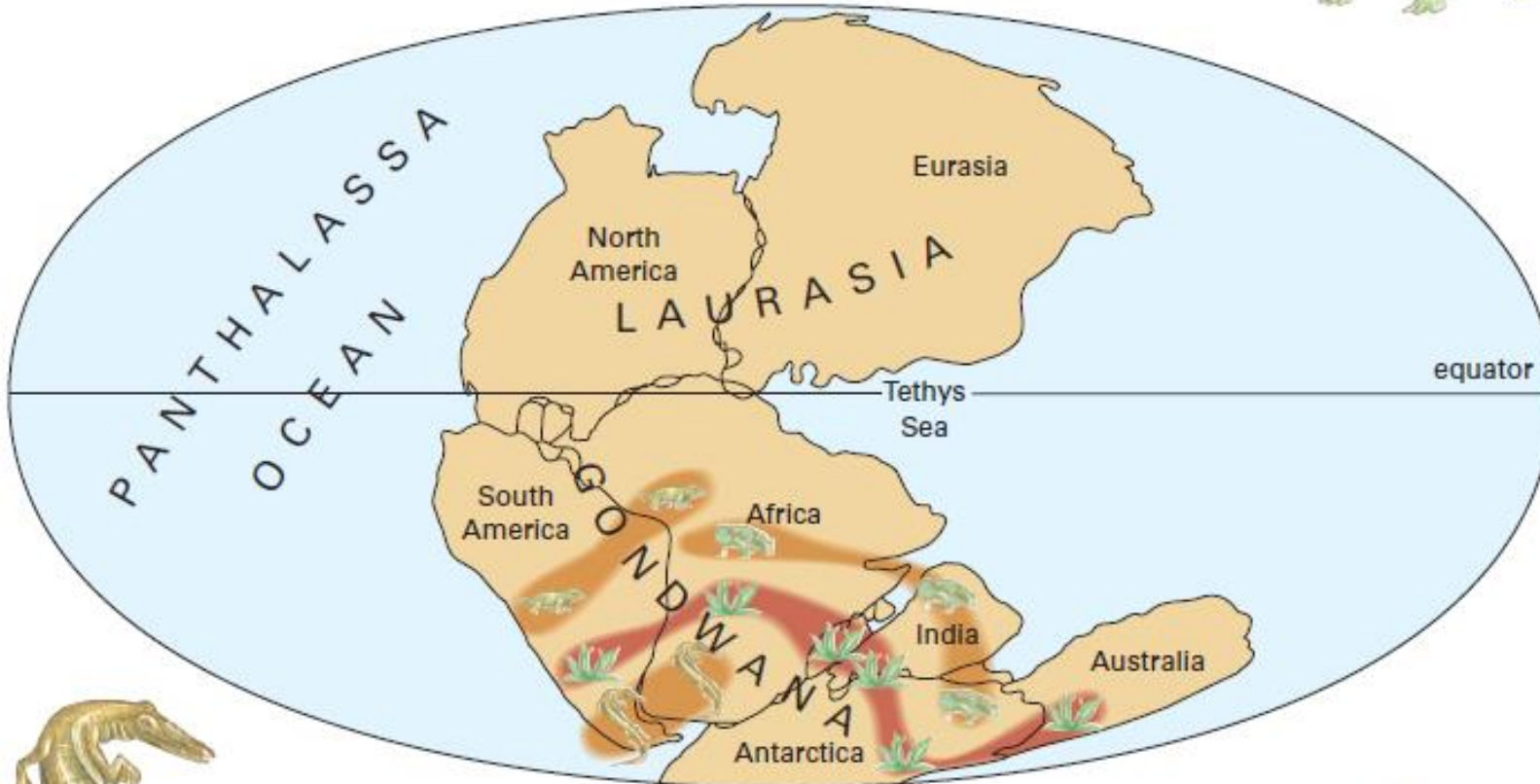


Continental drift theory – Matching features



Fossil evidence tells us that *Cynognathus*, a Triassic reptile, lived in Brazil and Africa.

Remains of *Lystrosaurus* were found in Africa, Antarctica, and India.



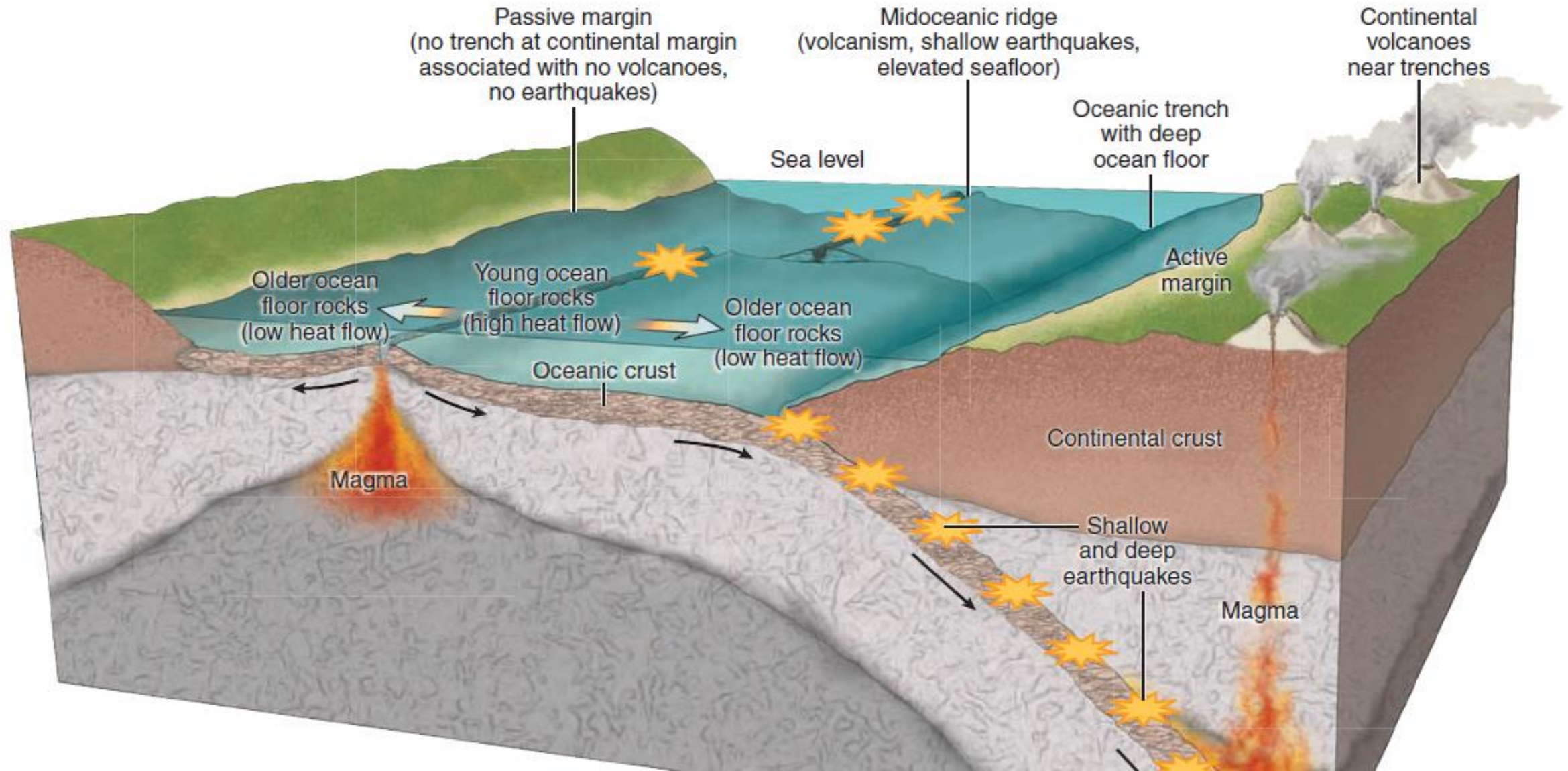
Wegener noted that fossils of *Mesosaurus* were found in Argentina and Africa but nowhere else in the world.

The fossil fern *Glossopteris* was found in all the southern land masses.

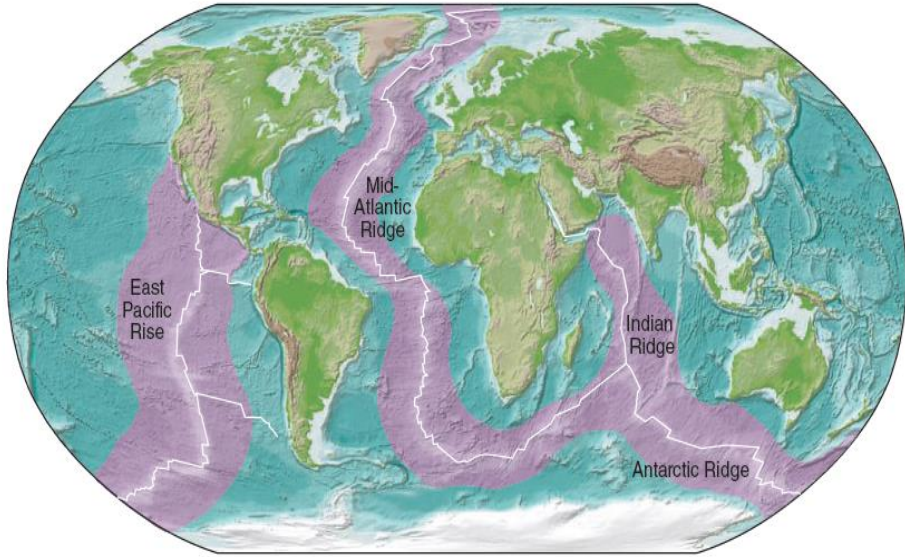


How did the continents move ?

First Proof of the concept - The Search for a Mechanism (Seafloor spreading)



Ocean ridge



- The oceanic ridge system consists of broad areas of shallower seafloor
- There are four principal regions: the Mid-Atlantic Ridge, the East Pacific Rise, the Antarctic Ridge, and the Indian Ridge, and many smaller connecting sections.
- The ridge system is a source of volcanic activity.



First Proof of the concept – The Search for a Mechanism

Age of the ocean floor

Analysis of rock samples from the ocean floor has revealed that the oceanic crust (< 200 million years) is relatively young compared to the age of the continents (> 4000 million years).

Seafloor spreading theory suggested that new oceanic lithosphere (crust and upper mantle) is being *continuously* formed along the ridge system by magma rising from below, and as this occurs, the existing rocks of the seafloor move away from the ridge.



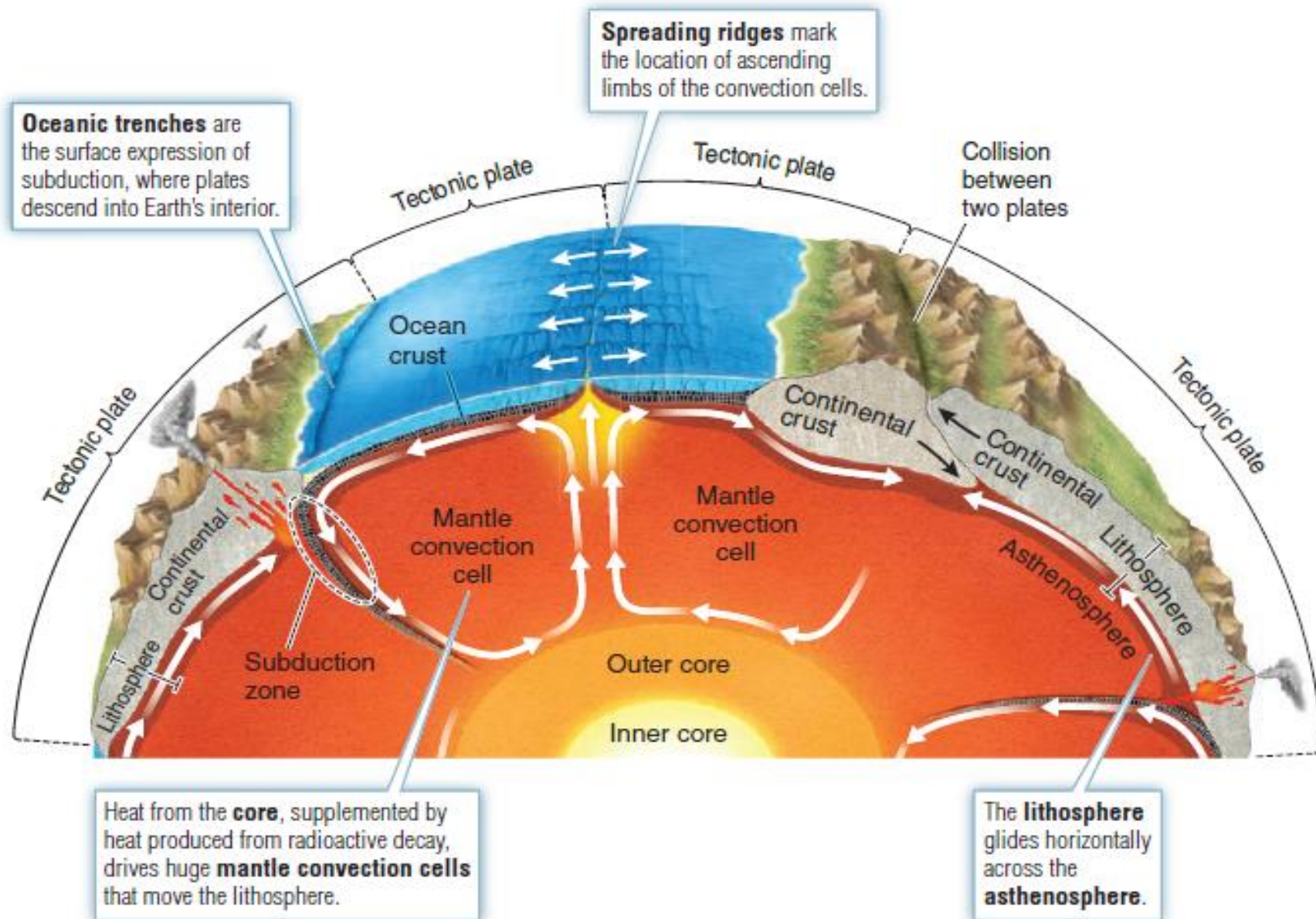
Convection as driving force

Hess and Dietz proposed convection

- Rising limbs of convecting mantle are present under midocean ridges.
- The upwelling mantle accounts for high heat flow observed under the mid-ocean ridges.
- The entire ocean is “virtually swept clean” as it is replaced by new ocean floor every 300–400 million years.
- The continents are carried passively along by the convecting mantle.
- The ocean floor and its thin cover of sediments, as well as all volcanic seamounts and guyots also ride down into the “jaw crusher” of the descending limb of a mantle convection cell (subduction).

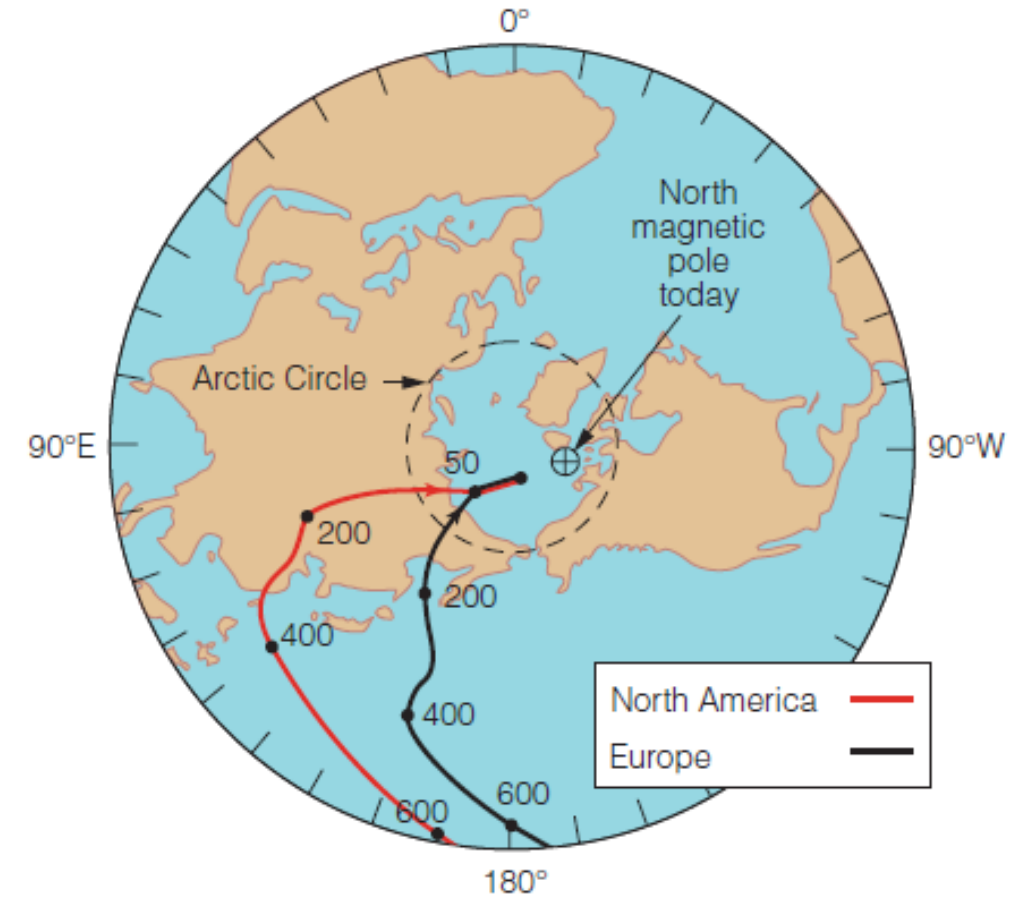
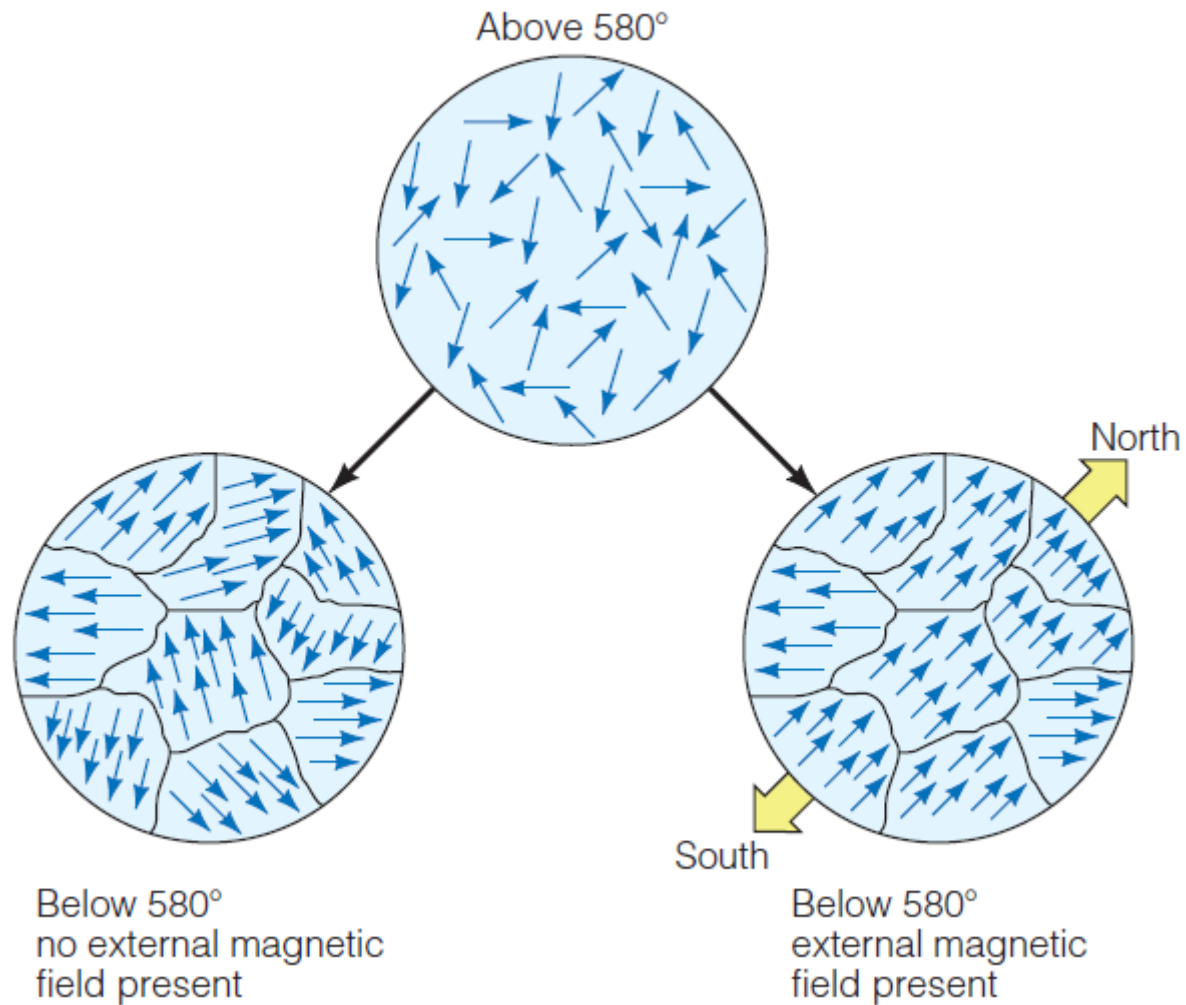


Will Earth expand because of this addition of new material ?



Second Proof of the concept – Apparent Polar Wandering

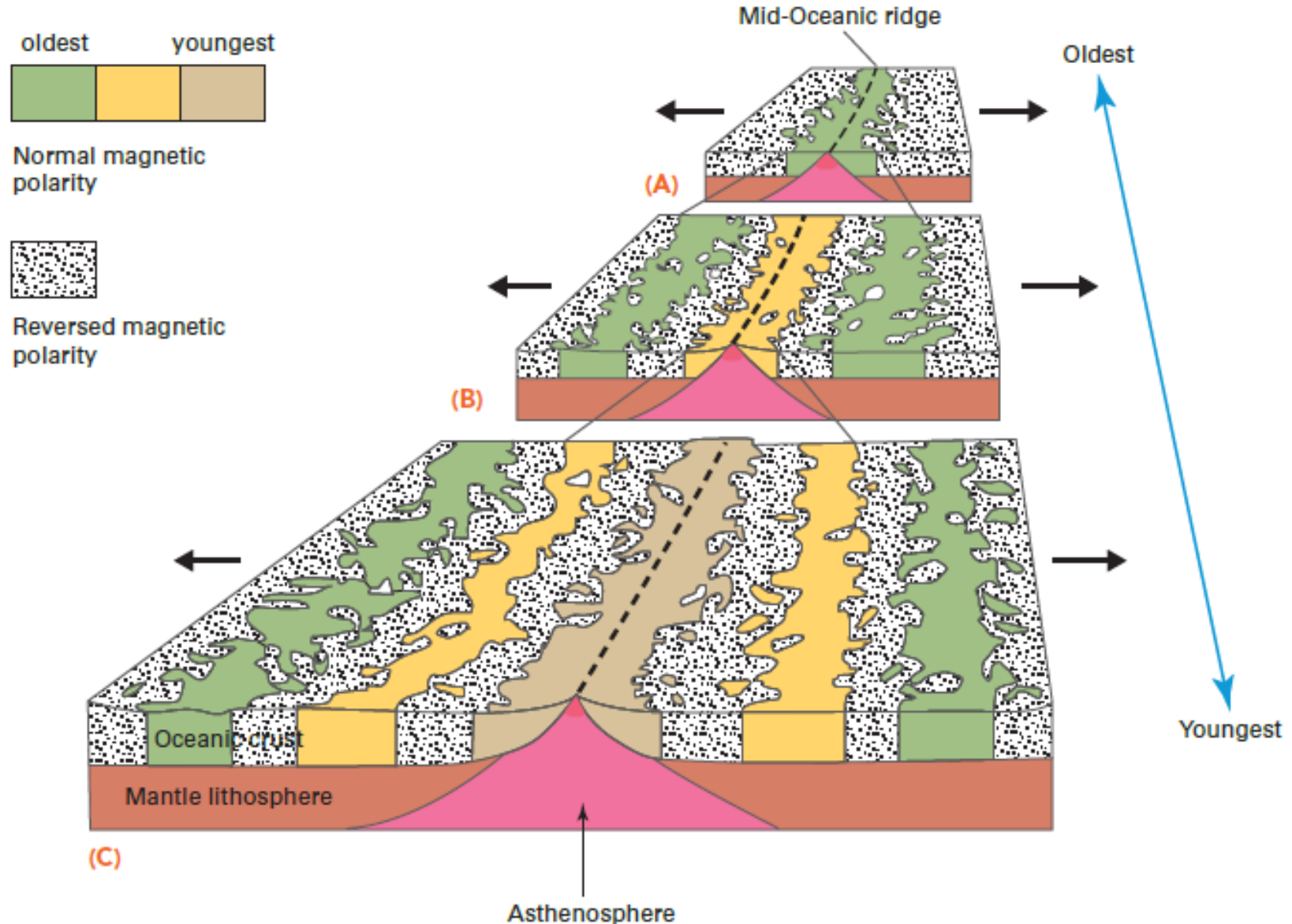
Paleomagnetism



Apparent Polar Wandering

How Paleomagnetism Supports the Seafloor Spreading Theory

As new oceanic crust cools at the Mid-Oceanic Ridge, it acquires the magnetic orientation of Earth's field. Alternating stripes of normal (colored) and reversed (black stippled pattern) polarity record reversals in Earth's magnetic field that occurred as the crust spread outward from the ridge. The three frames show the evolution of the spreading center through time from oldest (A) to youngest (C).



Theory of Plate tectonics

- It states that the lithosphere is a shell of hard, strong rock about 100 kilometers thick that floats on the hot, plastic asthenosphere the lithosphere is broken into seven large (and several smaller) segments called *tectonic plates*.
- Lithosphere is composed of several large, thin, relatively rigid “**plates**” that move relative to one another.

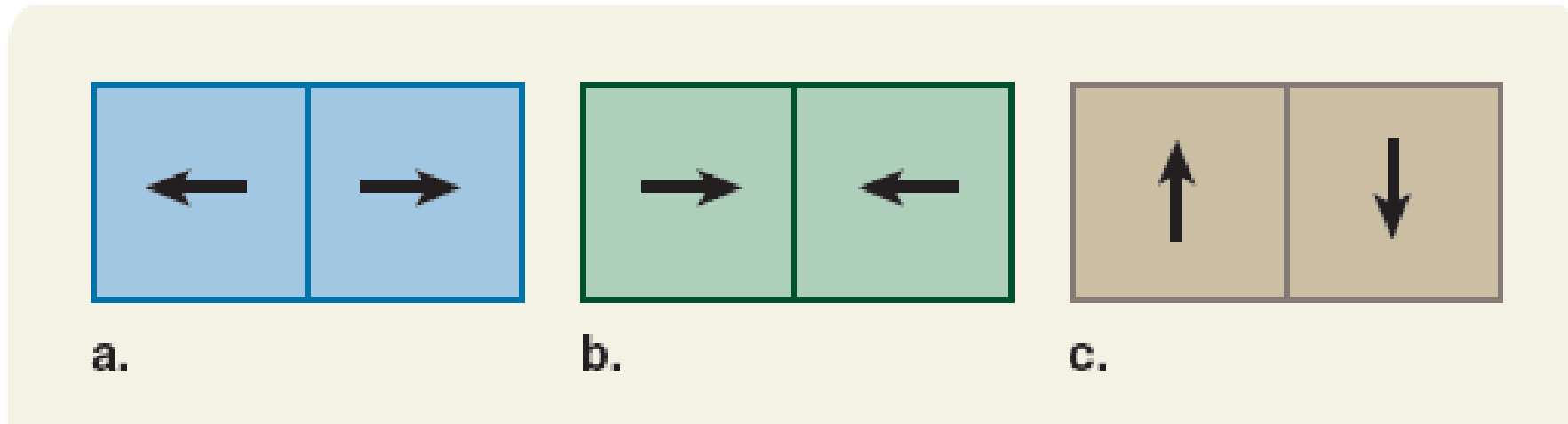
The theory of plate tectonics states that interactions of the plates along these boundaries account for the formation of new lithosphere, mountains, earthquakes, and volcanoes and contribute to the gradual movement of continents and the opening of oceans

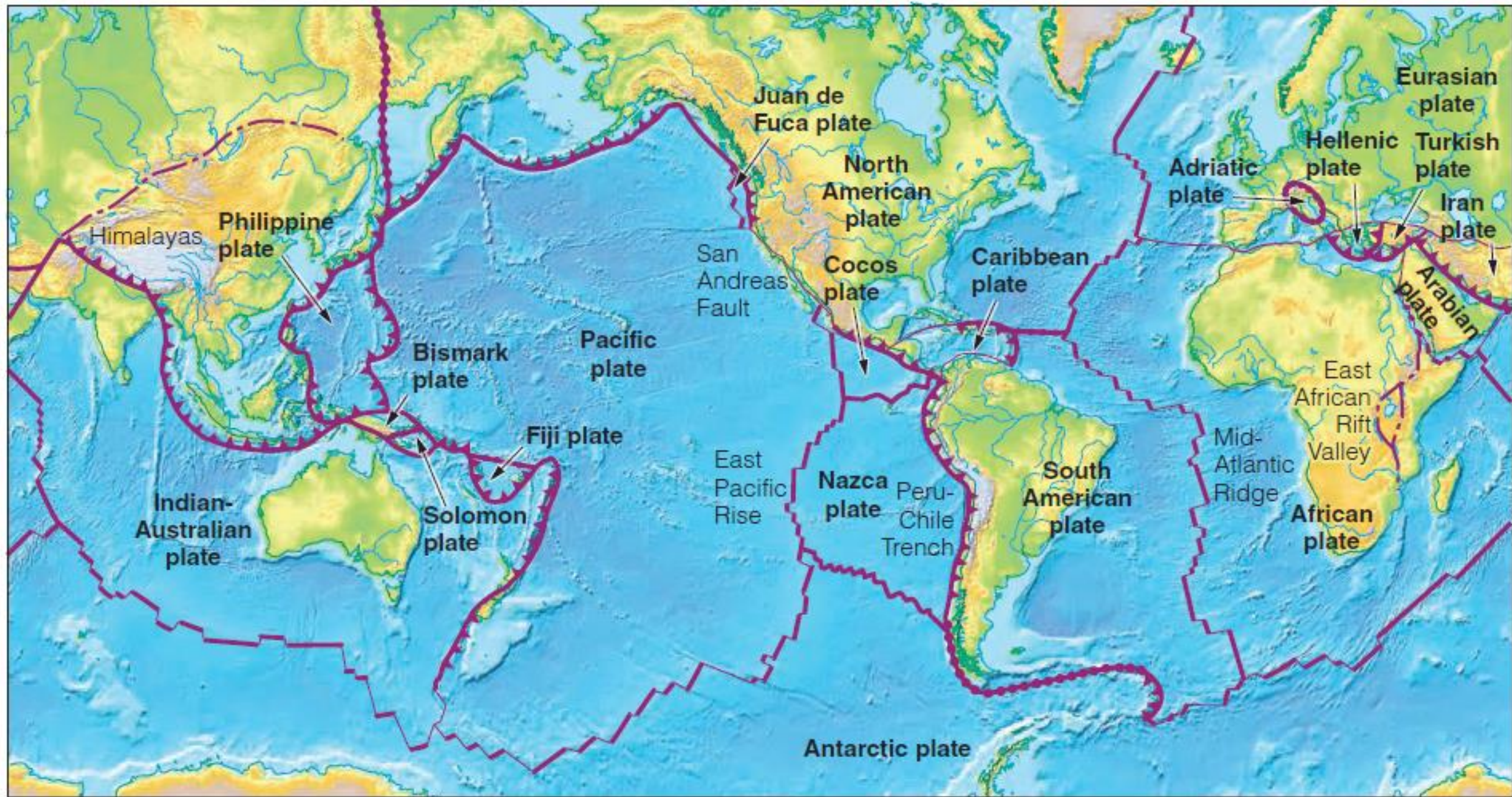


Plate boundaries

A **plate boundary** is a fracture that separates one plate from another. Neighboring plates can move relative to one another at these boundaries in three ways, shown in Figure.

- a) Divergent plate boundary
- b) Convergent plate boundary
- c) Transform plate boundary





Ridge axis

Divergent boundary



Transform fault

Transform boundary



Subduction zone

Convergent boundary



Zones of extension within continents



Uncertain plate boundary

Active Volcanoes, Plate Tectonics, and the "Ring of Fire"



Crust

Continental crust

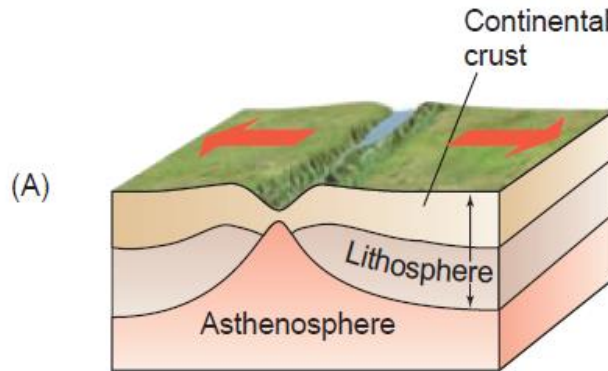
- Thicker up to 70 of km
- Less dense materials (granite)
- Cannot be subducted into mantle.
- Generally older

Ocean crust

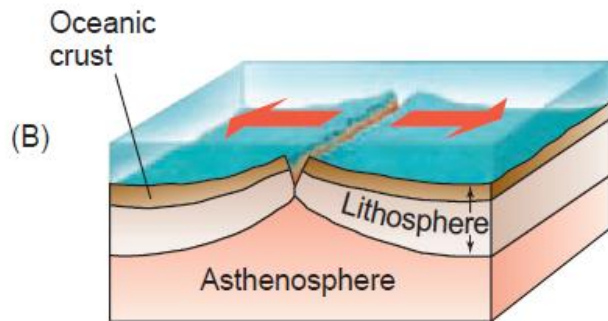
- Thinner (< 10 km)
- More dense materials (basalt)
- Can be subducted into mantle.
- Gets recycled continuously



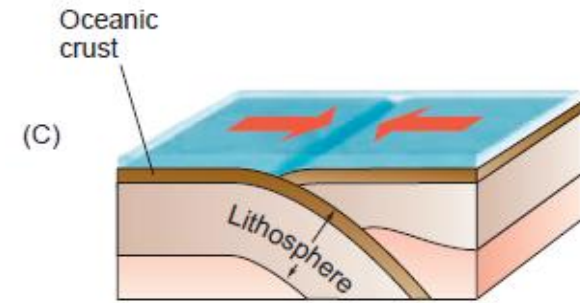
Plate boundaries



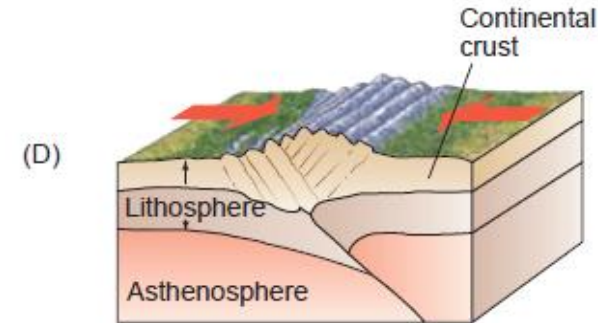
Continental divergent margin (continental rift)



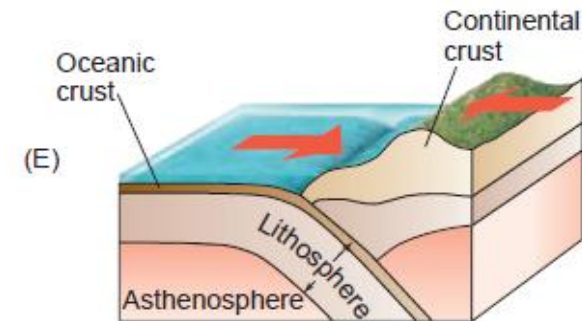
Oceanic divergent margin (oceanic rift)



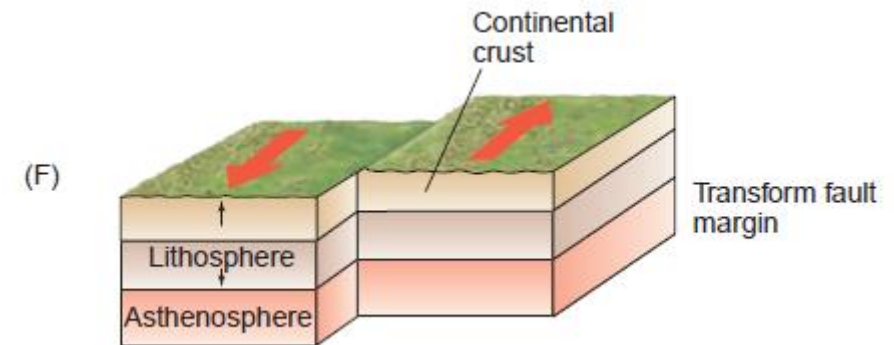
Ocean-ocean convergent margin (subduction zone)



Continent-continent convergent margin (collision zone)



Continent-ocean convergent margin (subduction zone)



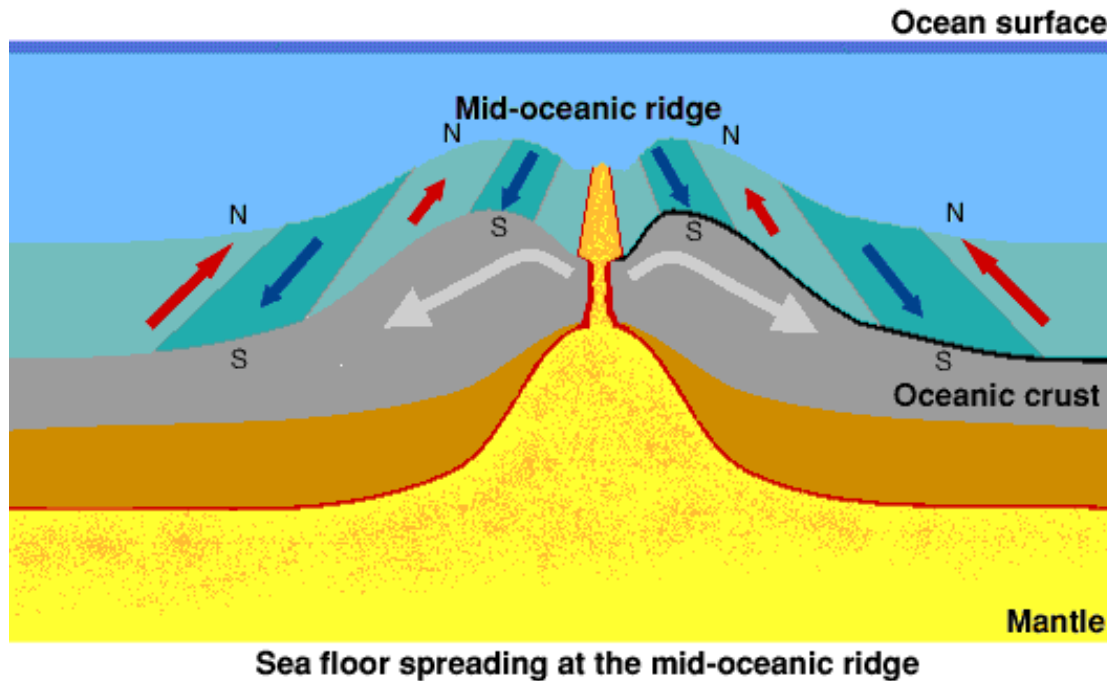
Transform fault margin

Lithospheric plates interact along plate boundaries in three principal ways, as shown by the arrows on this diagram

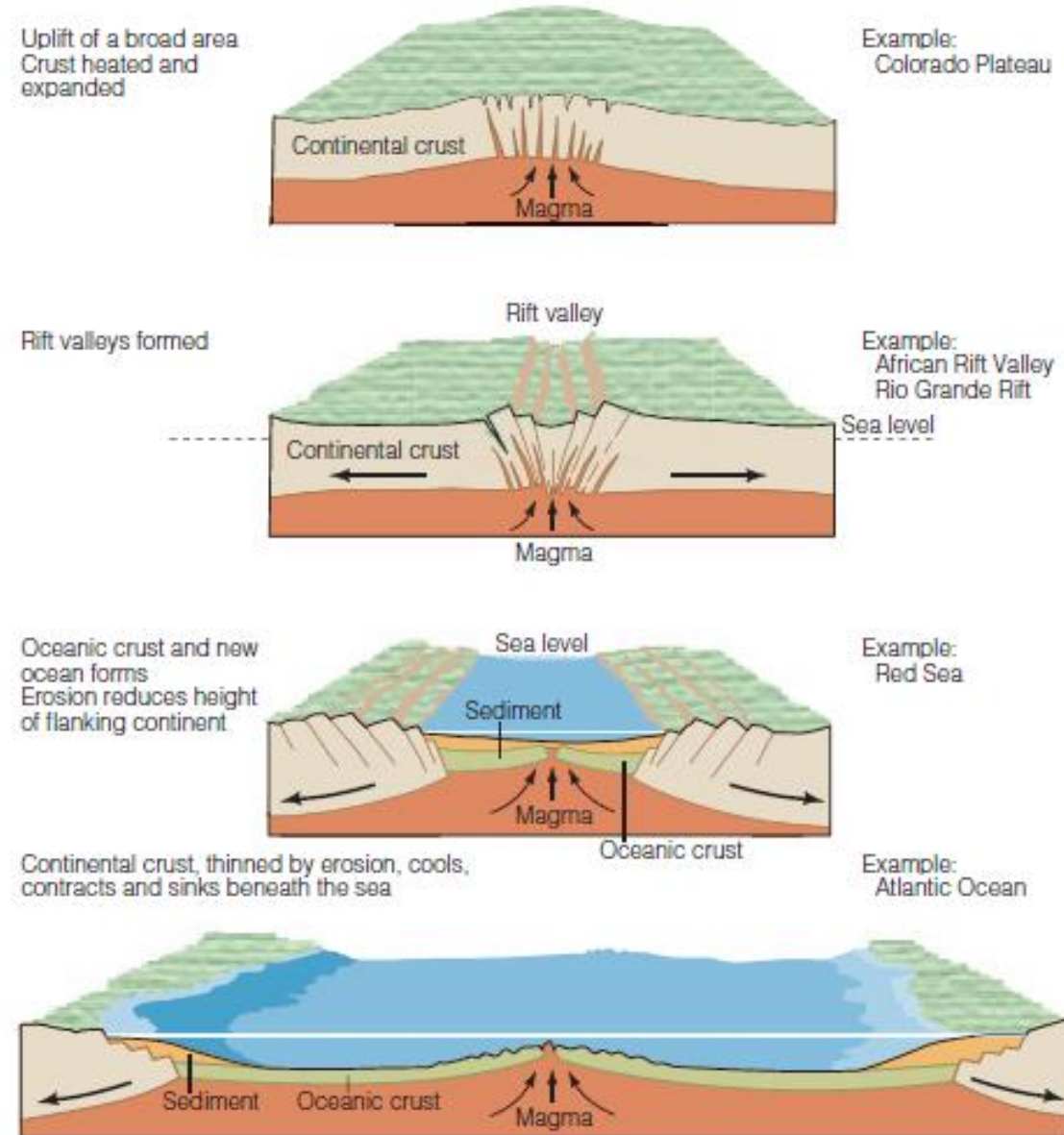


Divergent plate boundaries

Divergent plate boundaries are locations where tensional forces are pulling things apart. In locations where lithospheric plates are diverging, the rates of divergent motion range from 2 to 17 cm/year.



→ direction of Earth's magnetic field (recorded in solidified lava) ←



Convergent plate boundaries

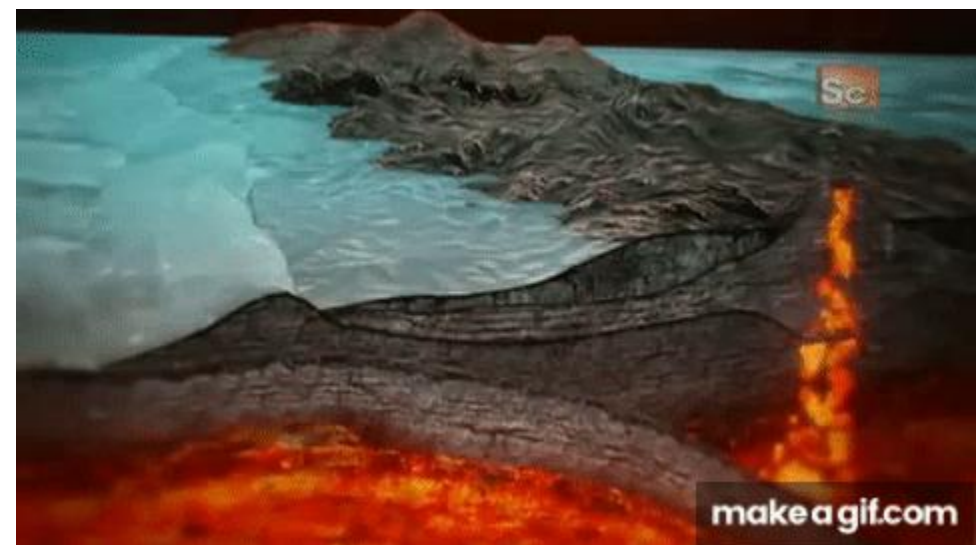
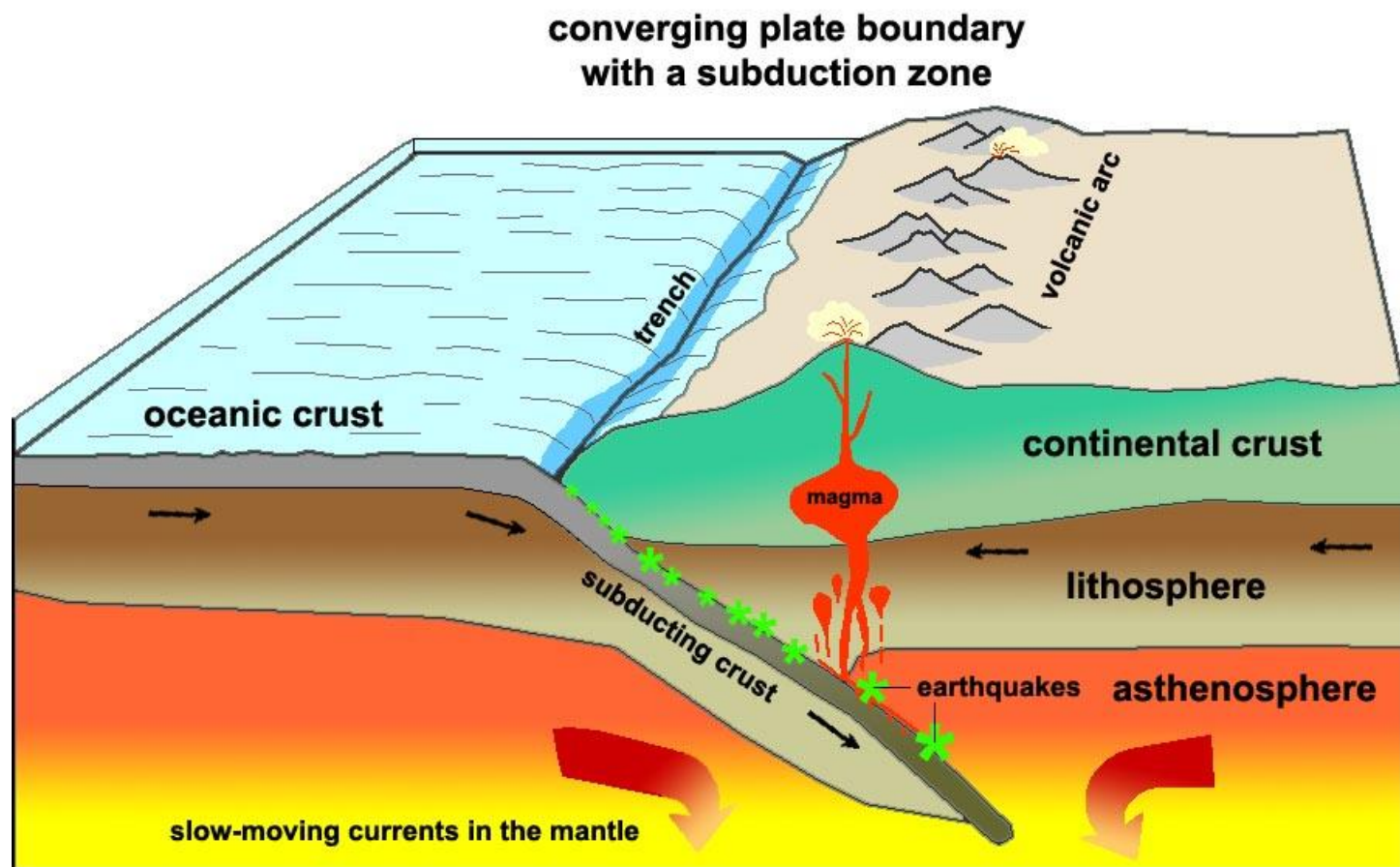
- Three types of Convergent Plate Boundaries: OC/CC, OC/OC, & CC/CC
- Three types of convergent plate boundaries are recognized: .
 - a) Subduction of ocean crust (OC) beneath continental crust (CC)
 - b) Subduction of ocean crust (OC) beneath ocean crust (OC)
 - c) Continental Collisions: continental crust (CC) colliding with continental crust (CC).

Most **tsunamis** are generated by subduction-zone-related earthquakes.



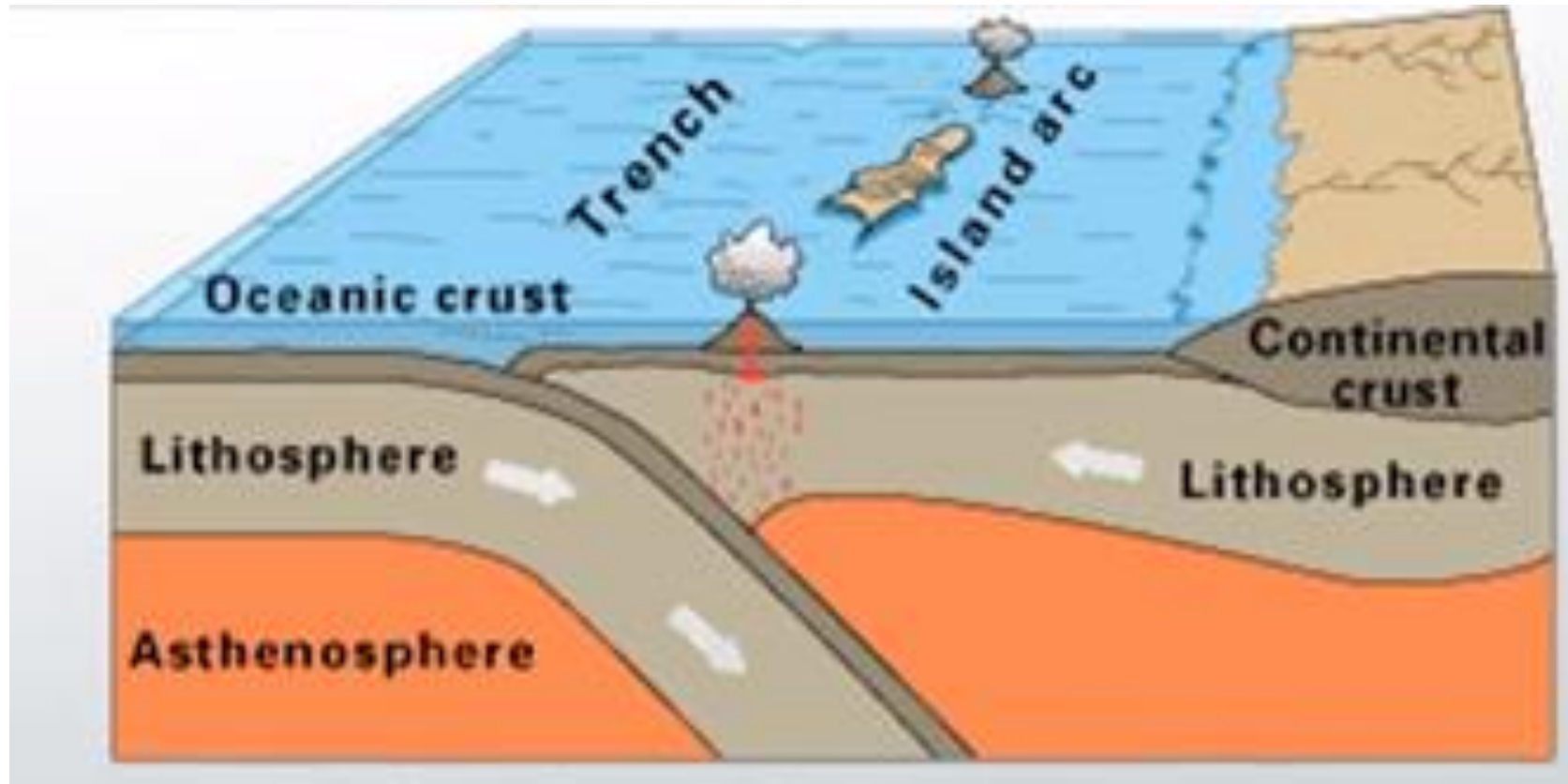
Subduction of ocean crust (OC) beneath continental crust (CC)

- Rates of subduction are up to 15 cm/yr in the active margins of the Pacific Basin.
- Examples: Andes in South America, Cascades in United States (include such volcanoes as Mt. St. Helens, Mt. Rainier, Mount Shasta, Crater Lake and many others)



Subduction of ocean crust (OC) beneath ocean crust (OC)

- Many similar features as above [OC/CC].
- Denser, older, cooler OC is pushed or subducted beneath less dense, warmer, younger OC.
- Volcanoes not as explosive as above with OC/CC, as there is no mixing of CC rocks (called granites). Volcanic rocks are mostly basaltic in composition.
- Examples: Japan, Tonga Islands, and Aleutian Islands (Alaska)



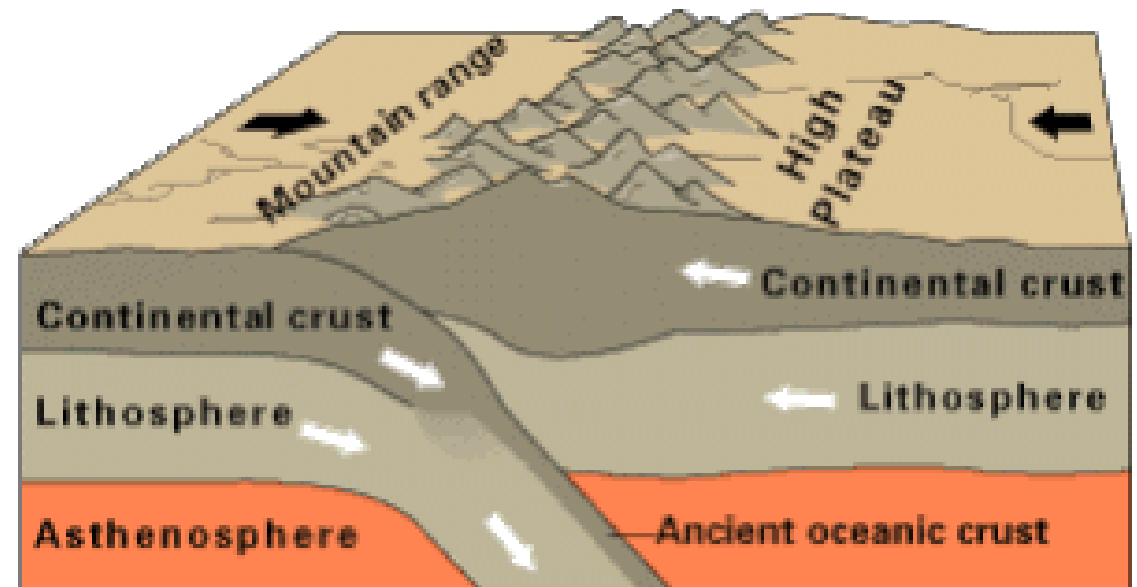
Continental Collisions: continental crust (CC) colliding with continental crust (CC)

When continents collide with other continental landmasses:

- Neither of the CC are subducted,
- Mountain building occurs with lots of earthquakes; massive erosion also occurs.

Examples

- Himalayas (India) beginning 45 million years ago)
- Alps Mountains are being pushed up by collisions between Africa (and Italian Peninsula) with Europe.
- Appalachians Mountains in the eastern United States (formed when North America collided with Africa about 350-400 million years ago (before the Atlantic Ocean opened later).



Continental-continental convergence

Transform plate boundaries

Transform boundaries are locations where one plate is sliding past another.

- Can occur in any crustal type (OC or CC)
- Crust is neither produced or destroyed

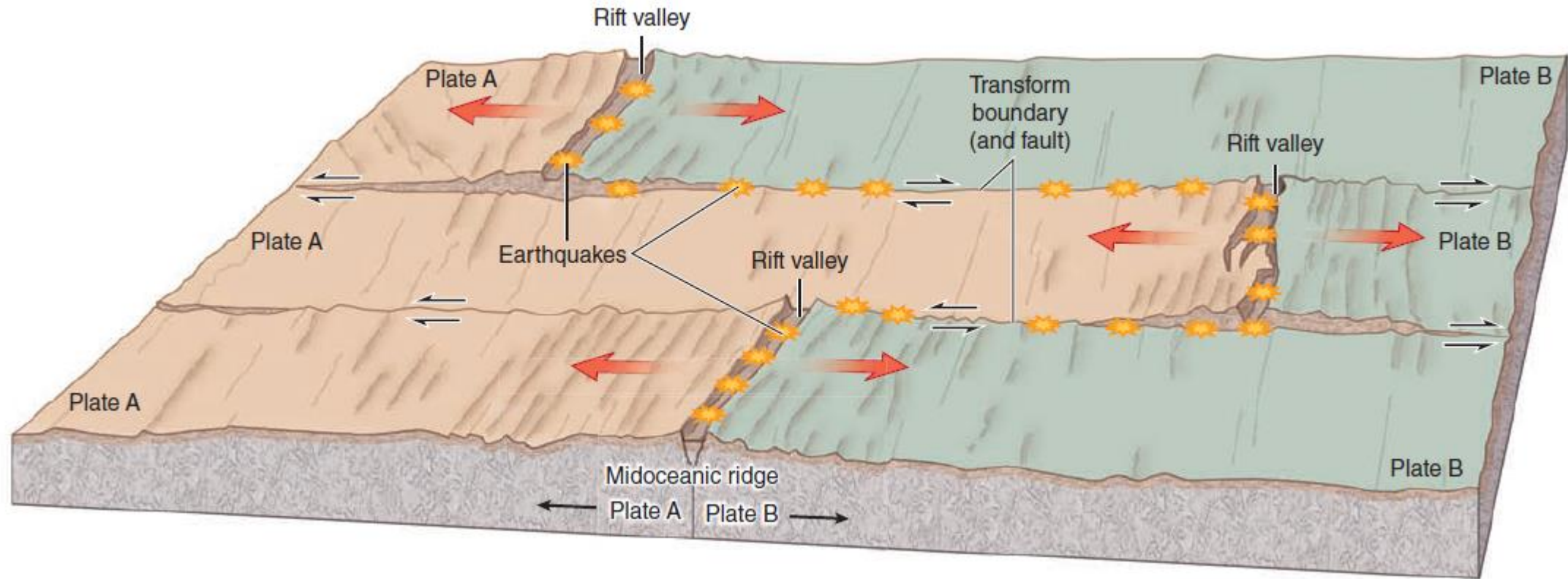
Oceanic Transform Faults

- In ocean crust, smaller transform faults occur perpendicular to the spreading centers.
- They have smaller earthquakes associated with them, when compared with their continental cousins.
- These faults occur where there is a bend or change in the rate of spreading along the MOR.

Continental Transform Faults

- In continental crust, transform faults can be large destructive faults like the San Andreas Fault System in California • Earthquake magnitudes up to 8.5 have been measured.
- Rates on the San Andreas Fault (SAF) average 2-5 cm/yr or LA will be adjacent to SF in about 20 million years!
- Compression(and uplift) or tension (and down-warping) can occur. The high mountains in Southern California are a result of this kind of compression.

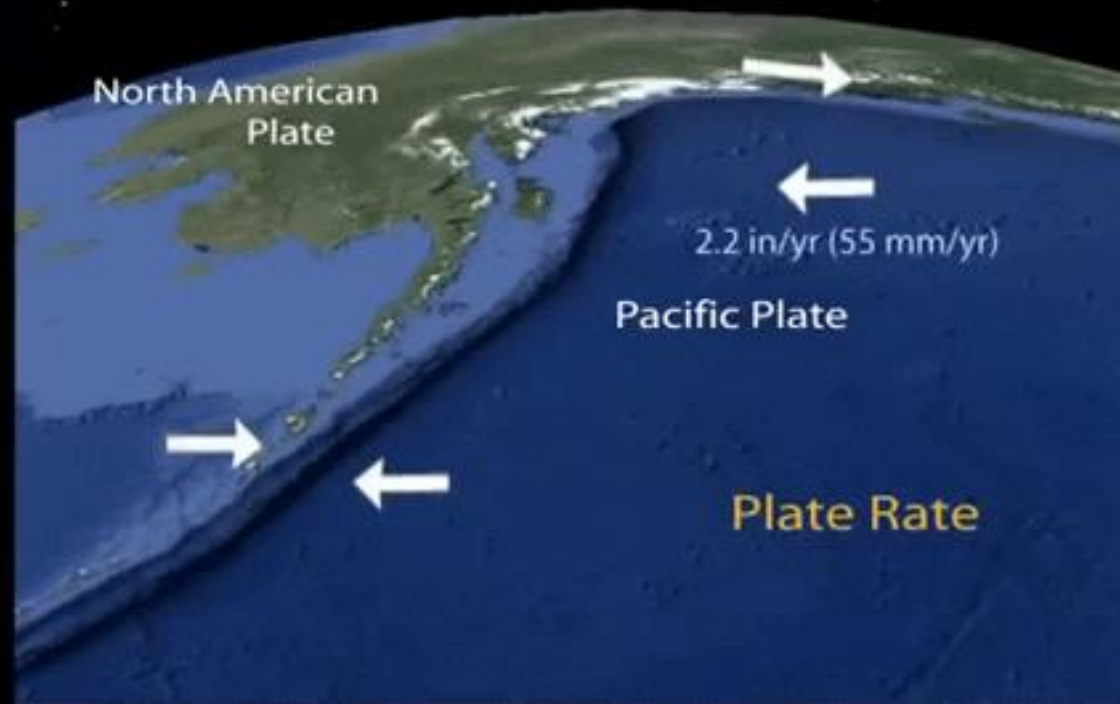




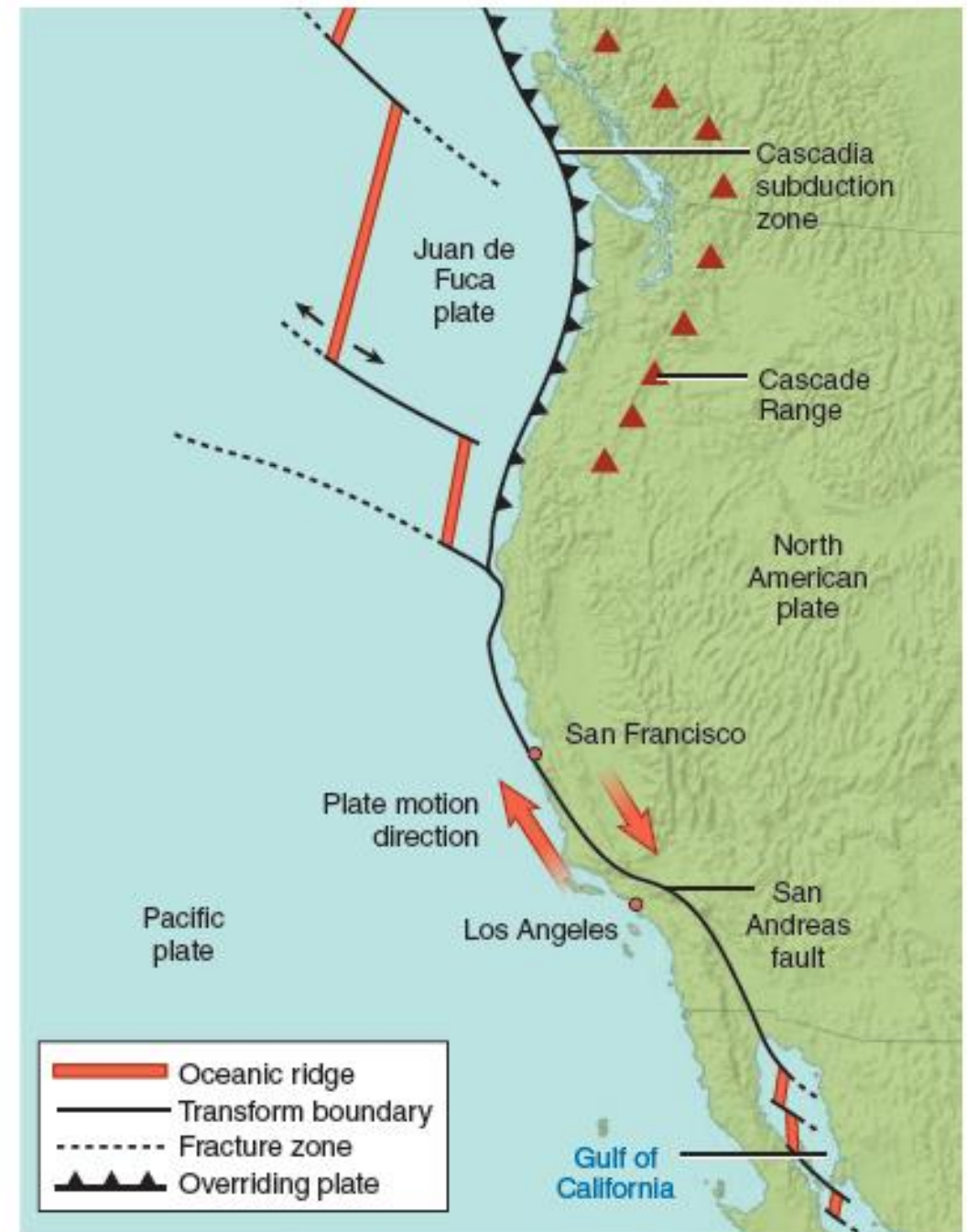
Features of transform boundaries that offset divergent boundaries. Plates A (tan) and B (green) move in opposite directions on either side of the transform boundary, causing earthquakes (yellow).



Alaska's major tectonic boundaries



San Andreas fault. The San Andreas fault is a transform boundary that separates the North American and Pacific plates. The smaller Juan de Fuca plate lies between these two plates opposite Oregon, Washington, northern California, and part of British Columbia. The Pacific plate moves northwest relative to the North American plate. Over the next several million years, Los Angeles will migrate toward San Francisco.



Characteristics and Examples of Plate Boundaries

Type of Boundary	Types of Plates Involved	Topography	Geologic Events	Modern Examples
Divergent	Ocean–ocean	Mid-Oceanic Ridge	Seafloor spreading, shallow earthquakes, rising magma, volcanoes	Mid-Atlantic Ridge
	Continent–continent	Rift valley	Continents torn apart, earthquakes, rising magma, volcanoes	East African Rift
Convergent	Ocean–ocean	Island arcs and ocean trenches	Subduction, deep earthquakes, rising magma, volcanoes, rock deformation	Western Aleutians
	Ocean–continent	Mountains and ocean trenches	Subduction, deep earthquakes, rising magma, volcanoes, rock deformation	Andes
	Continent–continent	Mountains	Deep earthquakes, rock deformation	Himalayas
Transform	Ocean–ocean	Major offset of Mid-Oceanic Ridge axis	Earthquakes	Offset of East Pacific Rise
	Ocean–continent	Linear, deformed mountain ranges	Earthquakes, rock deformation	Northern portion of San Andreas Fault
	Continent–continent	Linear, deformed mountain ranges	Earthquakes, rock deformation	Southern San Andreas Fault

Shallow earthquakes are between 0 and 70 km deep; intermediate earthquakes, 70 – 300 km deep; and deep earthquakes, 300 – 700 km deep

Rate of Plate Movements.

- The plate speeds are calculated based on satellite measurements and from paleomagnetic measurements.
- The close agreement between different measurement techniques implies that plates tend to move steadily rather than by starts and stops.
 - **Ultralow**: < 1 cm/year (Arctic ocean)
 - **Slow**: 1 to 2 cm/year (North America and Eurasian plates)
 - **Rapid**: >15 cm/year (Pacific and Nazca plates)
- Plates that carry lots of continental crust, such as the African, North American, and Eurasian plates, move relatively slowly.
- In contrast, plates without any continental crust, such as the Pacific and Nazca plates, move relatively rapidly; in fact, these are the fastest moving plates.

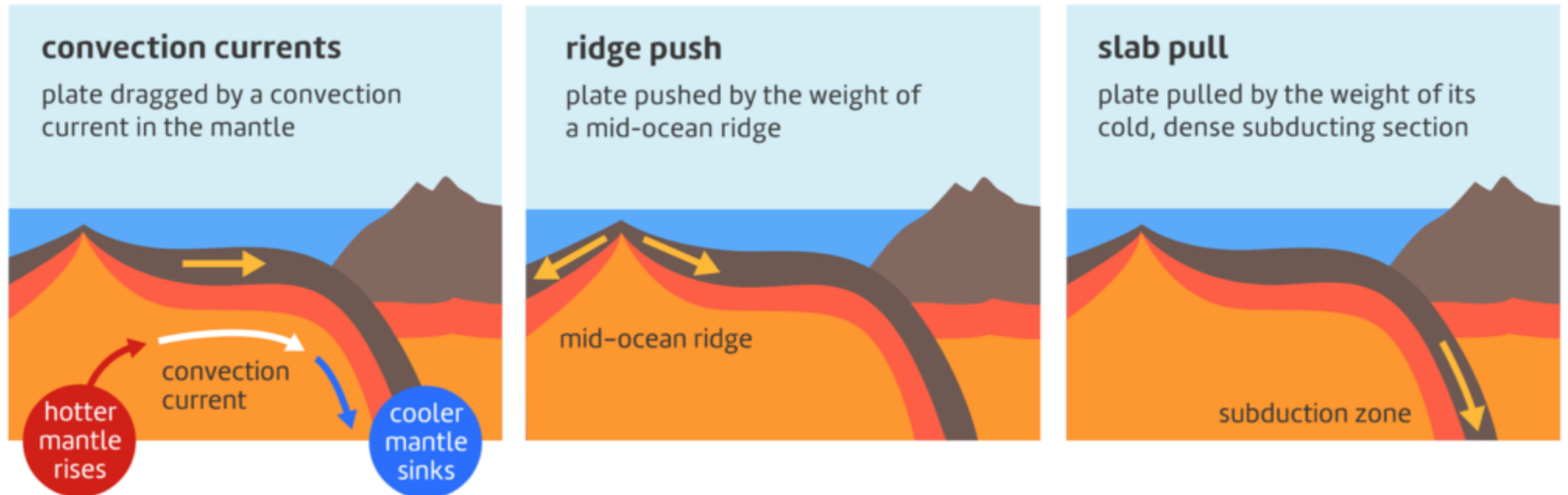


What drives plate motions?

- Researchers agree that convective flow in the mantle is the basic driving force of plate tectonics.

Forces that drive plate motion:

- Slab-pull - the slow pull of mantle material where it moves from a rising location to a sinking location.
- Slab-suction - The high density of cold ocean crust sinking into the mantle pulls crust with it.
- Ridge-push - new ocean crust is warm and tends to rise above the ocean floor, pushing older cooler crust away



How does this affect the sizes of lithospheric plates?

- A plate that has a spreading ridge but no subduction zone must grow in size; the African and North American plates are examples.
- To keep things in balance, plates with subduction zones must be slowly shrinking.
- Most modern subduction zones are located around the edge of the Pacific Plate; thus, much of the oceanic lithosphere now being destroyed is in the Pacific.
- The Indian Ocean, the Atlantic Ocean, and most other oceans are growing larger, while the Pacific Ocean is steadily getting smaller.
- It is estimated that about 200 million years in the future the Pacific Ocean will have disappeared and Asia and North America will have collided as a result.



Do Other Planets Have Plate Tectonics ?

- Mars cooled more rapidly than Earth and lost their **internal heat** much earlier in their history.
- There is no evidence that plate tectonics is occurring today on Mars, but surveys reveal that the oldest sections of the Martian crust have paleomagnetic patterns similar to those of Earth's ocean floors.
- The shape and size of volcanoes on Venus suggest it has a thick lithosphere that hinders the motion of plates.
- Without the destruction of old lithosphere, there is no opportunity to create new lithosphere and thus to allow the planet's internal heat to escape along long-lived stable features such as oceanic ridges.

Why does the Moon have craters ?

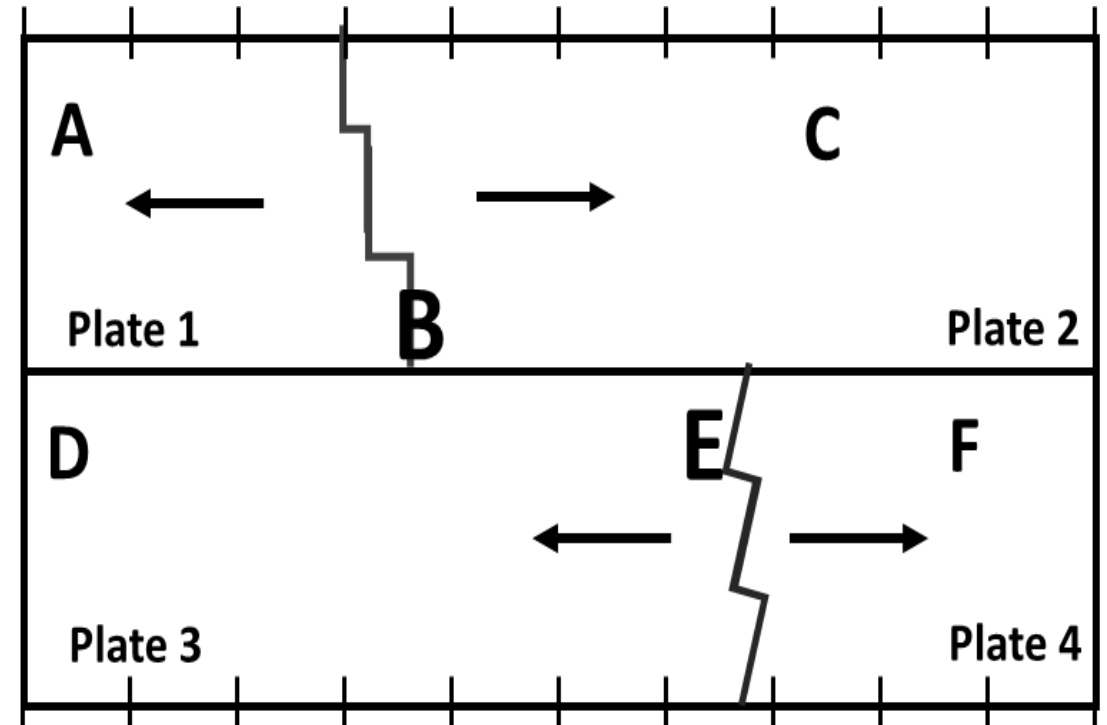
<https://www.youtube.com/shorts/O96MiXxXz6k>



Questions

The interactions of tectonic plates beneath an ocean are shown below, with arrows indicating the direction the crust is moving. Answer the below questions in one sentence.

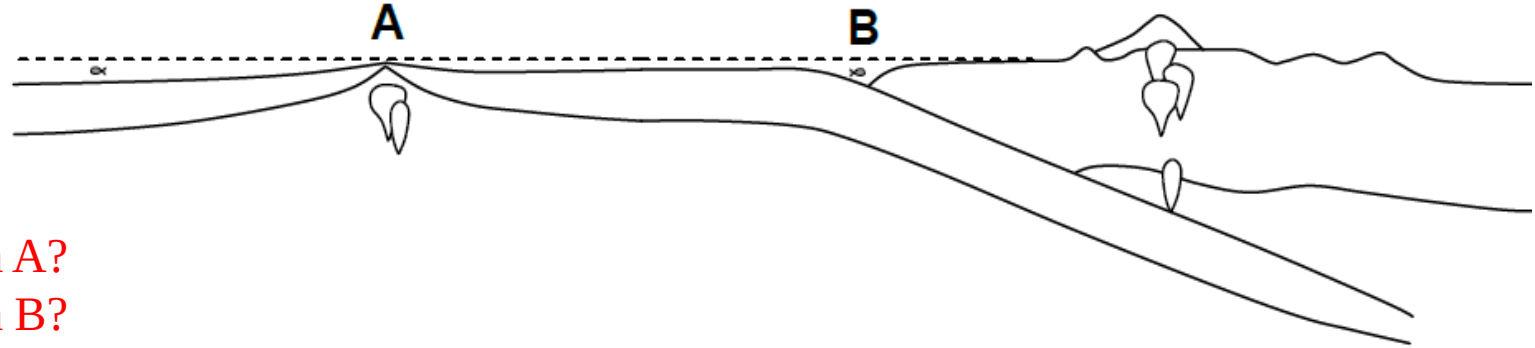
- Identify the type of plate boundary interactions between Plates 1, 2, and 3 as shown in the figure.
- At which location, and on which plate, can we find the oldest crust?
- If each plate is moving at a rate of 2 cm per year, roughly how many years did it take for Rocks C to reach their current locations?
- Name the geological events that occur between Plates 2 and 4, and between Plates 1 and 2.
- Where can we find the younger crust, and what type of topography is formed at those locations?



(The box is 10 cm wide)

Questions

The cross-section below shows the tectonic plates beneath an ocean and nearby continent. The sea level is indicated by the dashed line. Answer the below questions in one sentence.



- What type of plate boundary is at location A?
- What type of plate boundary is at location B?
- At which location can we observe subduction?
- What kind of topography is formed at location B on the continental plate?
- What geological events occur at location A?
- At which location can we find deep oceanic trenches?
- Where can we find the younger crust?
- Deep-focus earthquakes occur along or near which location?
- What drives the plate motions?
- Which of these plates has greater density?

